MATH 202 Your name: _____

Learning target D1, version 1

Check whether a function is a solution to a differential equation, and explain why the function is or is not a solution.

Consider the differential equation

$$\frac{dh}{dt} = -2\sqrt{h}.$$

Check whether each of the following is or is not a solution.

1. $h(t) = t^2 + 4t + 4$ (Hint: Factor.)

2. $h(t) = (10 - t)^2$

3. $h(t) = 4e^{2t}$